Thus it follows that we have the following statistical distance:

$$
\begin{aligned}
B_{F,\mathcal{M}}[m_{\theta_1} : m_{\theta_2}] &:= F(\theta_1) + F^*(\eta_2) - \theta_1^\top \eta_2, && (161)\\
&= -h(m_{\theta_1}) - \int p_0(x) \log m_{\theta_2}(x) \mathrm{d}\mu(x) - \int \sum_i \theta_1(i) p_i(x) \log m_{\theta_2}(x) \mathrm{d}\mu(x) \\
&\quad + \sum_i \theta_1(i) p_0(x) \log m_{\theta_2}(x) \mathrm{d}\mu(x), && (162)\\
&= -h(m_{\theta_1}) - \int ((1 - \sum_i \theta_1(i)) p_0(x) + \sum_i \theta_1(i) p_i(x)) \log m_{\theta_2}(x) \mathrm{d}\mu(x), && (163)\\
&= -h(m_{\theta_1}) - \int m_{\theta_1}(x) \log m_{\theta_2}(x) \mathrm{d}\mu(x), && (164)\\
&= \int m_{\theta_1}(x) \log \frac{m_{\theta_1}(x)}{m_{\theta_2}(x)} \mathrm{d}\mu(x), && (165)\\
&= D_{\mathrm{KL}}[m_{\theta_1} : m_{\theta_2}]. && (166)
\end{aligned}
$$

Thus we have $D_{\mathrm{KL}}[m_{\theta_1} : m_{\theta_2}] = B_F(\theta_1 : \theta_2)$. By relaxing the mixture densities m_{θ_1} and m_{θ_2} to arbitrary densities m_1 and m_2, we find that the dually flat geometry induced by the negentropy of densities of a mixture family induces a statistical distance which corresponds to the (forward) KL divergence. That is, we have recovered the statistical distance D_{KL} from $B_{F,\mathcal{M}}$. Note that in general the entropy of a mixture is not available in closed-form (because of the log sum term), except when the component distributions have pairwise disjoint supports. This latter case includes the case of Dirac distributions whose mixtures represent the categorical distributions.

Dually flat spaces can be built from any strictly convex C^3 generator F. Vinberg and Koszul [120] showed how to obtain such a convex generator for homogeneous cones. A cone $\mathcal{C}$ in a vector space V yields a dual cone of positive linear functionals in the dual vector space V^*:

$$\mathcal{C}^* := \{\omega \in V^* : \forall v \in \mathcal{C}, \omega(v) \geq 0\}. \tag{167}$$

The characteristic function of the cone is defined by

$$\chi_{\mathcal{C}}(\theta) := \int_{\mathcal{C}^*} \exp(-\omega(\theta)) \mathrm{d}\omega \geq 0, \tag{168}$$

and the function $\log \chi_{\mathcal{C}}(\theta)$ defines a Bregman generator which induces a Hessian structure and a dually flat space.

Figure 10 displays the main types of information manifolds encountered in information geometry with their relationships.

4 Some applications of information geometry

Information geometry [8] found broad applications in information sciences. For example, we can mention:

- Statistics: Asymptotic inference, Expectation-Maximization (EM and the novel information-geometric em), time series (AutoRegressive Moving Average model, ARMA) models,
- Machine learning: Restricted Boltzmann machines (RBMs), neuromanifolds and natural gradient [124],
- Signal processing: Principal Component Analysis (PCA), Independent Component Analysis (ICA), Non-negative Matrix Factorization (NMF),
- Mathematical programming: Barrier function of interior point methods,
- Game theory: Score functions.

Next, we shall describe a few applications, starting with the celebrated natural gradient descent.

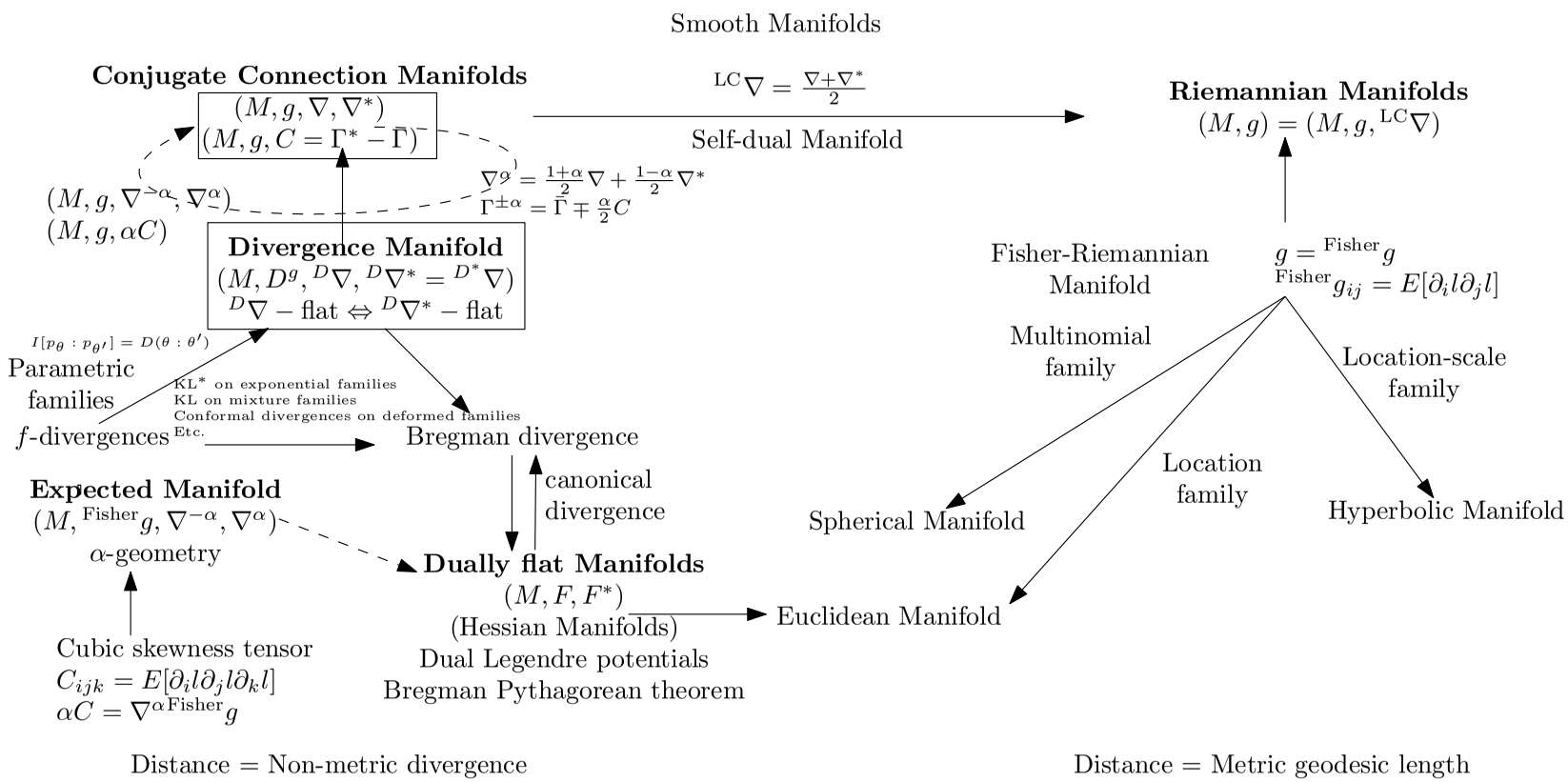

Figure 10: Overview of the main types of information manifolds with their relationships in information geometry.

4.1 Natural gradient in Riemannian space

The *Natural Gradient* [6] (NG) is an extension of the ordinary (Cartesian) gradient of Euclidean geometry to the gradient in a Riemannian space analyzed in an arbitrary coordinate system. We explain the natural gradient

4.1.1 The vanilla gradient descent method

Given a real-valued function $L_\theta(\theta)$ parameterized by a a D-dimensional vector θ on parameter space $\theta \in \Theta \subset \mathbb{R}^D$, we wish to minimize L_θ, i.e., solve $\min_{\theta\in\Theta} L_\theta(\theta)$. The *gradient descent* (GD) method, also called the *steepest descent* method, is a first-order local optimization procedure which starts by initializing the parameter to an arbitrary value (say, $\theta_0 \in \Theta$), and then iteratively updates at stage t the current location of θ_t to θ_{t+1} as follows:

$$\text{GD}: \quad \theta_{t+1} = \theta_t - \alpha_t \nabla_\theta L_\theta(\theta_t). \tag{169}$$

The scalar $\alpha_t > 0$ is called the *step size* or *learning rate* in machine learning. The ordinary gradient (OG) $\nabla_\theta F_\theta(\theta)$ (vector of partial derivatives) represents the *steepest vector* at θ of the function graph $\mathcal{L}_\theta = \{(\theta, L_\theta(\theta)) \ : \ \theta \in \Theta\}$. The GD method was pioneered by Cauchy [28] (1847) and its convergence proof to a *stationary point* was first reported in Curry [35] (1944).

If we *reparameterize* the function L_θ using a one-to-one and onto differentiable mapping $\eta = \eta(\theta)$ (with reciprocal inverse mapping $\theta = \theta(\eta)$), the GD update rule transforms as:

$$\eta_{t+1} = \eta_t - \alpha_t \nabla_\eta L_\eta(\eta_t), \tag{170}$$

where

$$L_\eta(\eta) := L_\theta(\theta(\eta)). \tag{171}$$

Thus in general, the two gradient descent location sequences $\{\theta_t\}_t$ and $\{\eta_t\}_t$ (initialized at $\theta_0 = \theta(\eta_0)$ and $\eta_0 = \eta(\theta_0)$) are *different* (because usually $\eta(\theta) \neq \theta$), and the two GDs may potentially reach different stationary points. In other words, the GD local optimization *depends on the choice of the parameterization* of the function L (i.e., L_θ or L_η). For example, minimizing with the gradient descent a temperature function $L_\theta(\theta)$ with respect to Celsius degrees θ may yield a different result than minimizing the same temperature function $L_\eta(\eta) = L_\theta(\theta(\eta))$ expressed with respect to Fahrenheit degrees η. That is, the GD optimization is *extrinsic* since it depends on the choice of the parameterization of the function, and does not take into account the underlying geometry of the parameter space Θ.

The natural gradient precisely addresses this problem and solves it by choosing *intrinsically* the steepest direction with respect to a Riemannian metric tensor field on the parameter manifold. We shall explain the natural gradient descent method and highlight its connections with the Riemannian gradient descent, the mirror descent and even the ordinary gradient descent when the parameter space is dually flat.

4.1.2 Natural gradient and its connection with the Riemannian gradient

Let (M, g) be a D-dimensional Riemannian space [38] equipped with a metric tensor g, and $L \in C^\infty(M)$ a smooth function to minimize on the manifold M. The *Riemannian gradient* [21] uses the Riemannian *exponential map* $\exp_p : T_p \to M$ to update the sequence of points p_t's on the manifold as follows:

$$\text{RG}: \quad p_{t+1} = \exp_{p_t}(-\alpha_t \nabla_M L(p_t)), \tag{172}$$

where the Riemannian gradient ∇_M is defined according to a *directional derivative* ∇_v by:

$$\nabla_M L(p) := \nabla_v \left(L\left(\exp_p(v)\right)\right)\big|_{v=0}, \tag{173}$$

with

$$\nabla_v L(p) := \lim_{h\to 0} \frac{L(p+hv) - L(p)}{h}. \tag{174}$$